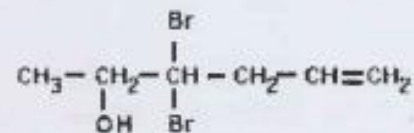


1.

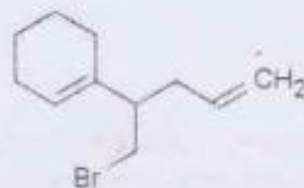
01.

(a). Give IUPAC names of the following compounds

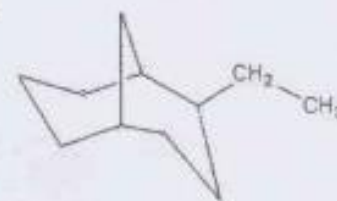
i).



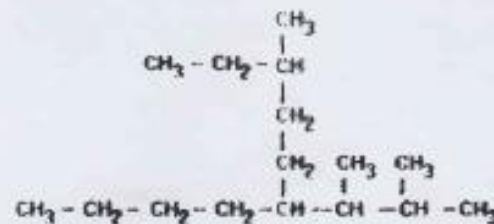
ii).



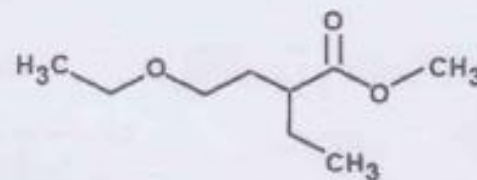
iii).



iv).



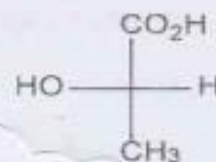
v).



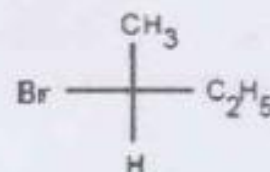
(10 marks)

(b). Give IUPAC names of the following compounds with R/S, *cis/trans* or E/Z configurations

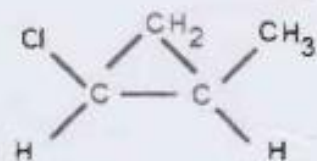
i).



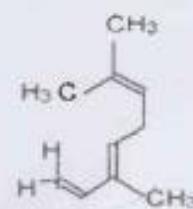
ii).



iii).



iv).



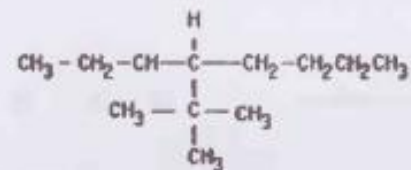
(12 marks)

(c). Draw the structure of the compound (Z, Z)-3,5-Octadienal.

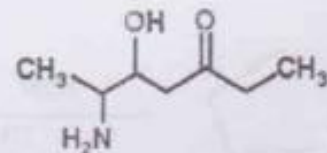
2.

(a). Give IUPAC names of the following compounds

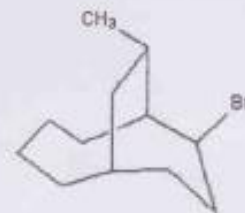
i).



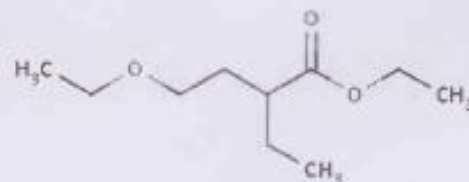
ii).



iii).



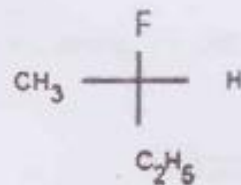
iv).



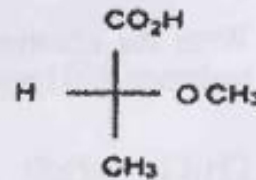
(12 marks)

(b). Give IUPAC names of the following compounds with R/S, *cis/trans* or E/Z configurations

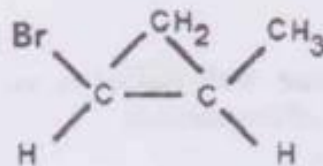
i).



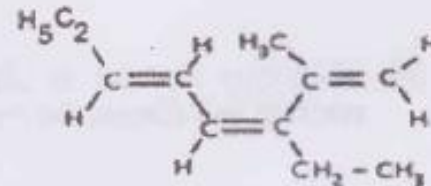
ii).



iii).



iv).



(12 marks)

(c). Draw the structure of the compound (2E, 4Z)-1-chloro-2,4-heptadiene

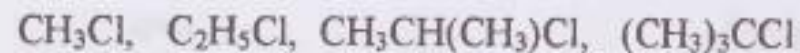
3.

(08 marks)

(b). Write a brief note on factors affecting S_N1 type reactions

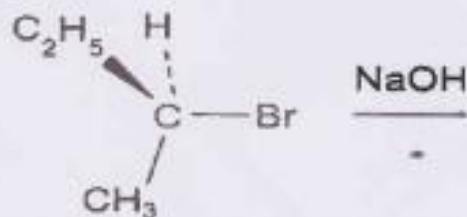
(06 marks)

(c). Write the equations and compare the rate of reactions when following compounds undergo S_N2 type reaction with HCN.



(04 marks)

(d). S_N2 type reaction of (S)-2-Bromobutane with NaOH is given below. Complete the reaction and discuss the energy diagram of the reaction.



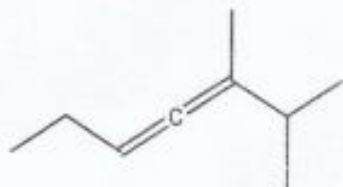
(07 marks)

01.

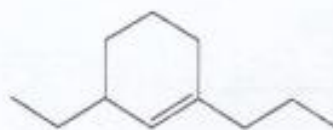
4.

(a). Give IUPAC names of the following compounds

i).



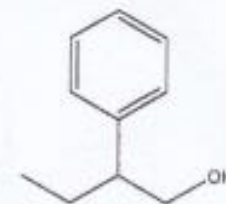
ii).



iii).



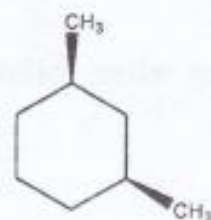
iv).



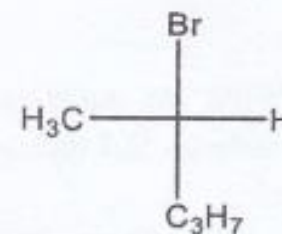
(10 marks)

(b). Give IUPAC names of the following compounds with R/S, *cis/trans* or E/Z configurations

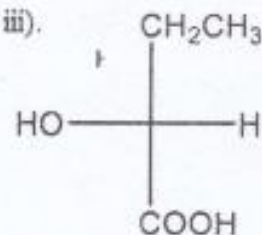
i).



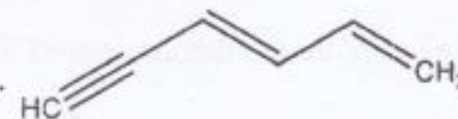
ii).



iii).



iv).



(12 marks)

(c). Draw the structure of the compound (2E, 4Z)-1-bromo-2,4-hexadiene.

(03 marks)

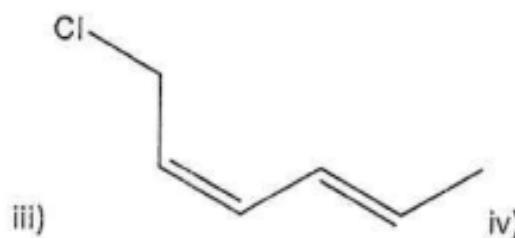
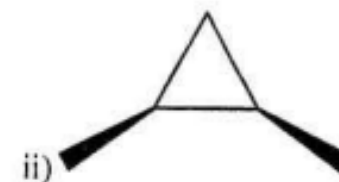
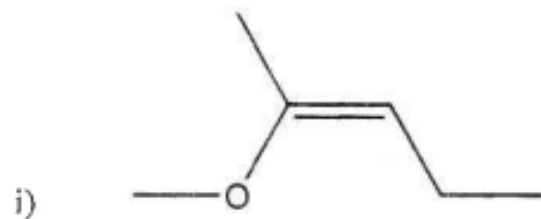
5.

01. (a). Draw the relevant structures for given IUPAC names.

- i) 3-Chloro-4-ethyl-6-nonene
- ii) 2,3,5-trimethyl-4-propylheptane
- iii) Bicycle-(2,2,1)-heptene
- iv) ethyl-3-hydroxyhexanoate

(10 marks)

(b). Give IUPAC names for the following compounds with R/S, *cis/trans* or E/Z configurations.



(12 marks)

(c). Draw the structure of the compound (R)-2-chloro-4-methylhept-3-ene

(03 marks)

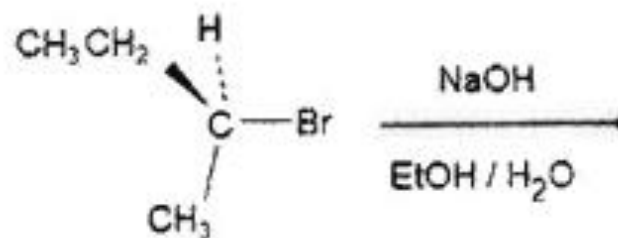
6.

02. (a). Compare the two mechanisms of nucleophilic substitution reaction.

(06 marks)

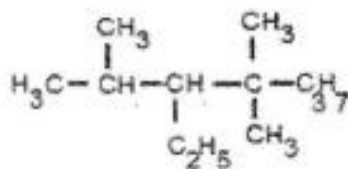
(b). Identify the intermediate and the final product, and discuss the stereoisomerism of the following reaction proceeds with S_N2 mechanism.

(07 marks)

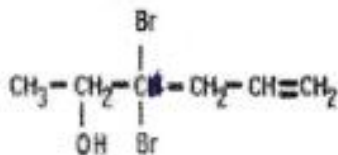


(S)-(+)-2-Bromobutane

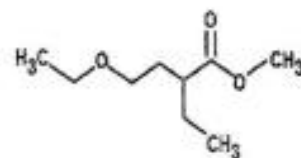
01. (a). Give IUPAC names of the following compounds.



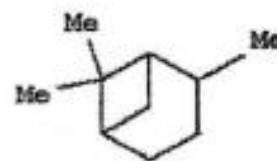
i



ii.



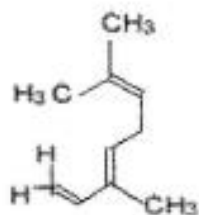
iii.



v.

(10 marks)

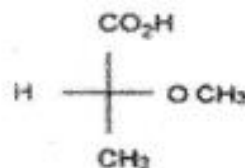
(b). Give IUPAC names of the following compounds with R/S, *cis/trans* or E/Z configurations.



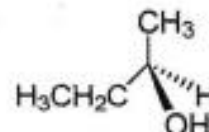
i



ii



iii



iv

(10 marks)

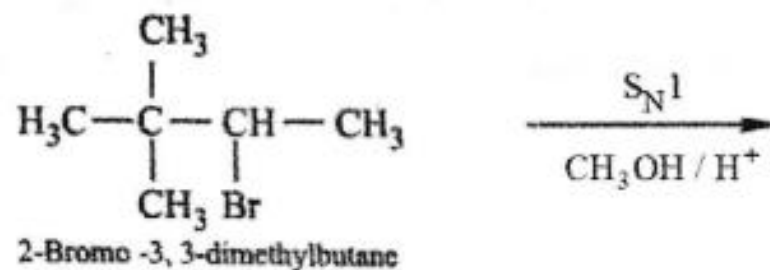
(c). Draw the structure of the following compounds

i. (S)-2-Chloro-4-methylhept-3-ene

ii) Bicyclo-(2,2,1)-heptene

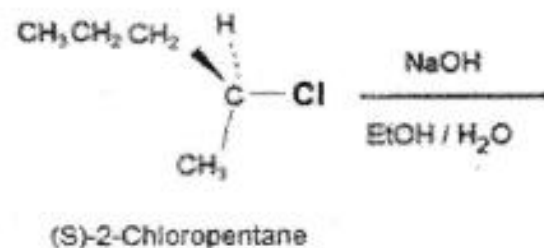
(05 marks)

(b). Write the mechanism of the following reaction



(06 marks)

(c). The given below is the $\text{S}_\text{N}2$ type reaction of (S)-2-Chloropentane with NaOH in ethanol-water.



i. Briefly discuss the factors that affect the rate of $\text{S}_\text{N}2$ type reaction. (04 marks)

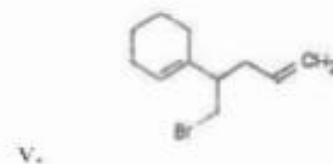
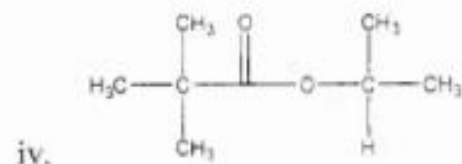
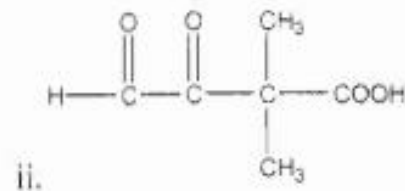
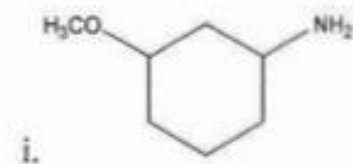
ii. Write all the steps and the mechanism of this reaction and draw the relevant energy diagram. (05 marks)

iii. Discuss the stereoisomerism of this reaction (04 marks)

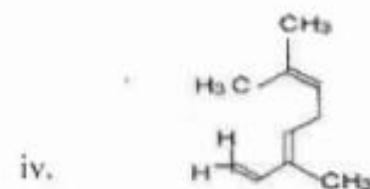
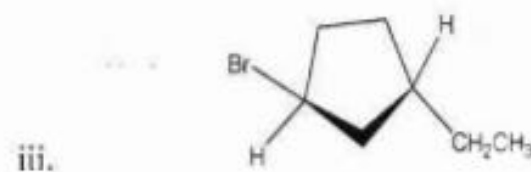
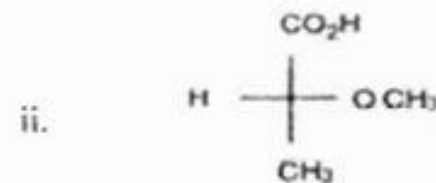
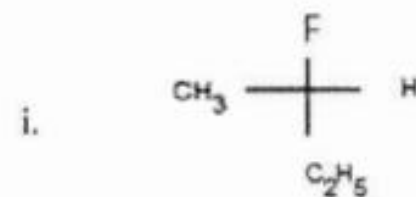
8.

01.

a). Give IUPAC names for the following compounds.



(10 marks)

b). Give IUPAC names for the following compounds with R/S, *cis/trans* or E/Z configurations

(12 marks)

c). Draw the structure of the compound (2E, 4Z)-1-Bromo-2,4-heptadiene

(12 marks)

9.

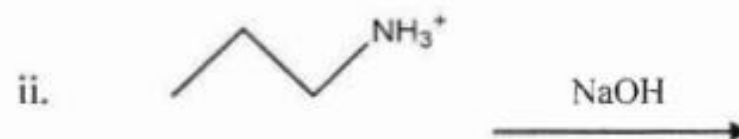
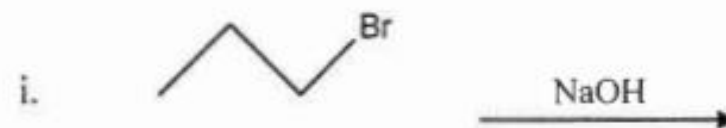
02.

a). Write a short note on effect of solvent with examples.

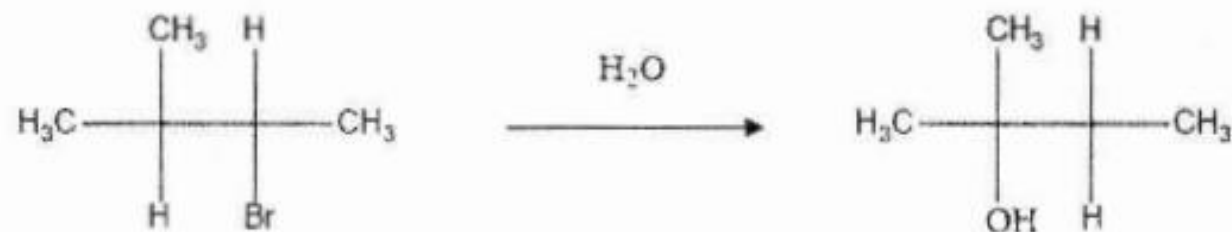
(05 marks)

b). Complete the following S_N1 reaction and compare the rates of reaction with each other.

(05 marks)



c). Write all the steps and the mechanism of this reaction



10.

- e). Write the equations and compare the rate of reactions when following compounds undergo S_N2 type reaction with HCN.



(04 marks)